

Candidate number _____

Score _____

2018 International Biology Olympiad National Team Selection Camp Practical Test Questions**First Trial Session A**

ÿ All necessary equipment and reagents for the experiment are on the table. Please check them against the list below. If anything is missing, please raise your voice.

Tell the judges with your hand.

Experimental equipment and reagents:

equipment	quantity
Microdispenser (P20/P200)	1 piece / 1 piece
Micropipette tips (2ÿL~200ÿL)	1 box
96-well micro-volume plate	1
1.5 mL microcentrifuge tubes/microcentrifuge tube	20 pieces/1 piece
rack (box),	1
timer, pen/white paper/	1 stick / 1 sheet / 1 bundle
ruler, simple calculator	1 unit
reagents	quantity
PBS	1 mL
BCA (Bicinchoninic acid), a protein chromogenic	4 mL
agent; Bovine serum albumin (BSA) standard (1.0 mg/mL);	100 ÿL
Acetaldehyde dehydrogenase (ALDH).	100 ÿL
Buffer A: 50 mM Tris-HCl buffer, pH 9, 0.1 M KCl, 10 mM ÿ-ME Buffer B: 50	0.5 mL
mM Phosphate buffer, pH 7, 0.1 M KCl, 10 mM ÿ-ME Buffer C: 50 mM Acetate	0.5 mL
buffer, pH 5, 0.1 M KCl, 10 mM ÿ-ME 1% ÿÿ (Acetaldehyde)	0.5 mL
	40 µL
NAD ⁺ (20 mM)	100 µL
ALDH reaction termination solution (1 mM EDTA)	0.5 mL
instrument	quantity
Microdisc spectrophotometer (common)	1 unit
Small benchtop centrifuge	1 unit

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. Please use public instruments in accordance with the instructions.
3. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
4. The total time allotted for answering questions (including practical questions A and B) is 90 minutes. Please answer the questions on this paper.

Practical Exercise A (50 points)

I. Experimental Topic: Determination of Protein Concentration and Enzyme Activity Analysis of Acetaldehyde Dehydrogenase

In alkaline copper sulfate solution, proteins can reduce copper ions from divalent to monovalent. Then, BCA (bicinchoninic acid) binds to the monovalent copper ions, turning the solution blue and exhibiting strong absorbance at 570 nm. The advantages of the BCA protein concentration analysis method are that BCA is more stable in alkaline solution than the Folin reagent used in the Biuret method, and it requires only one step.

After alcohol enters the body, it is oxidized to acetaldehyde by alcohol dehydrogenase in the liver. The acetaldehyde is then further converted to harmless acetic acid by aldehyde dehydrogenase. However, people who experience facial flushing after drinking alcohol have insufficient activity or gene deficiency in the aldehyde dehydrogenase, leading to a rapid accumulation of acetaldehyde in the body that cannot be metabolized quickly. This results in prolonged facial flushing, as the liver's P450 enzymes (a group of oxidases with relatively low specificity) slowly metabolize the ingested alcohol. Furthermore, acetaldehyde is more toxic than ethanol and is one of the main causes of hangovers. Aldehyde dehydrogenase (ALDH) uses nicotinamide adenine dinucleotide (NAD⁺) as a coenzyme to catalyze the oxidation of acetaldehyde to acetic acid. The catalytic reaction formula is as follows:

$\text{CH}_3\text{CHO} + \text{NAD}^+ \rightarrow \text{CH}_3\text{COOH} + \text{NADH} + \text{H}^+$. The enzyme reaction of alcohol dehydrogenase is usually measured by observing the change in absorbance at 340 nm when NAD⁺ is converted to NADH using a photometer.

II. Experiment Content:

1. Establish a regression line for BSA standards and determine the ALDH concentration for unknown protein concentrations.
2. Determine the optimal pH for ALDH.

III. Experimental Procedure: (10 points)

1. Please dilute the BSA standard with PBS and construct a standard regression line (see chart on the next page). When centrifuging the mixed solution using a small benchtop centrifuge, please pay close attention to equilibrium and safety! 2. Please note! Double replicates are required for the PBS blank control and each diluted standard concentration. The provided BSA standard will not be provided after it is used up, so please carefully calculate the required amount before starting the experiment.
3. The procedure for BCA protein quantification is as follows: Add 10 μL of PBS, BSA, or ALDH to a 96-well microplate, then add 200 μL of BCA solution for color development. After adding the color reagent, let it stand for 15 minutes before measuring the absorbance using a microplate spectrophotometer. The color development is extremely stable within 30 minutes; please follow the usage instructions and methods of the microplate spectrophotometer and do not rush. Air bubbles in the sample will affect the accuracy of the readings. ∇ Please use pipette tips sparingly; excessive waste will result in a deduction of points. ∇ Do not allow the micropipette tip to come into contact with the solution in the well when adding the color reagent to avoid contamination. ∇ The color reagent easily settles at the bottom; please gently tap the microplate to mix the added solution evenly. 4. Please prepare 500 μL of ALDH reaction matrix solution using the provided pH 9, pH 7, and pH 5 buffer solutions. The ALDH reaction matrix solutions for all three pH values must contain 0.01% acetaldehyde and 1 mM NAD⁺. 5. Please note! Each pH value reaction must be performed in duplicate. The provided acid-base buffer solutions and matrix will not be provided once used up; please calculate the required quantities carefully.
6. The method for analyzing the activity of ALDH is as follows: Add 10 μL of ALDH and 200 μL of self-prepared reaction matrix solutions with three different pH values to a 96-well microplate, and react at room temperature for 15 minutes. Then add 50 μL of reaction stop solution, and the absorbance can be measured using a microplate spectrophotometer.

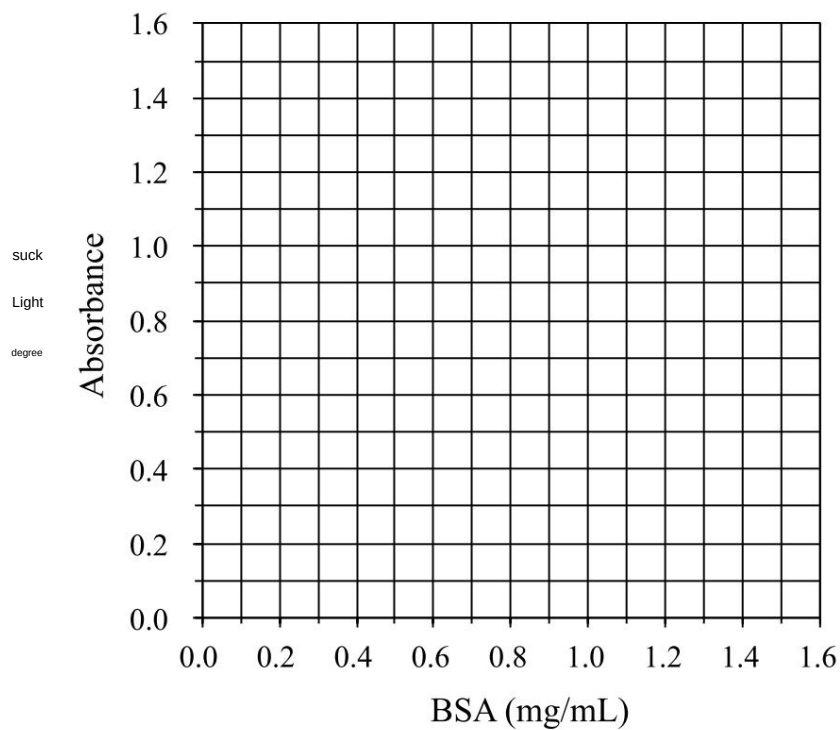
IV. Experimental Results and Questions Answered:

1. Please fill in the absorbance values for protein quantification in the blanks in the table below, and calculate the average absorbance value for each group. (7 points)

BSA concentration (mg/mL)	Reaction #1	Reaction #2	Average absorbance
0			
0.2			
0.4			
0.6			
0.8			
1.0			
ALDH			

2. Following on from the previous question, please mark the average absorbance value with a solid circle (•) in the following figure, and draw the line you estimated.

Sex regression line. (6 points)



3. Using the linear regression line in the above graph, what is the approximate concentration of ALDH for an unknown protein? Please answer.

Fill in the blanks in the boxes below. (5 points)

Estimated ALDH protein concentration	
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4. Fill in the absorbance values obtained from the ALDH enzyme activity analysis in the table below: (6 points)

pH	Reaction #1	Average absorbance of reaction #2	
9			
7			
5			

5. Based on the experimental results, what is the optimal pH value for ALDH? (Please tick the boxes in the table below) (4 points)

pH 9	pH 7	pH5

6. Please speculate on the role of NAD⁺ in the ALDH-catalyzed enzymatic reaction. (4 points)

7. Please speculate on why adding ALDH reaction stop solution can terminate the enzymatic reaction. (4 points)

8. Please speculate on the reason why γ -ME (γ -mercaptoethanol) needs to be added to the ALDH reaction matrix solution. (4 points)

The structure of γ -mercaptoethanol is shown on the right:

